

Four New Datasets: Consolidated Experimental Report

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Provenance

- Recorded mirror checksums: 17/17 matched.
- Primary-source verification: not completed. working copies are third-party mirrors; primary hosts were unavailable.
- Licence strings are inherited from the acquisition manifest and require primary-record confirmation.

Protocols

- Historical / Protocol A: preprocessing and PCA may see full data; retained only for comparability and explicitly leaky.
- Clean / Protocol B: 80/20 stratified split first; imputation, missing indicators, winsorization, encoding, constant removal, scaling, PCA and ADASYN fit on training only.
- PKDD transaction aggregates use the strict rule `transaction_date < loan_date`; undated standing orders are excluded.
- Experiment 27 uses the barcode-vector F_pq proxy, not true persistence-diagram distances.

Dataset audits

pkdd_czech

- Shape: 682 rows × 39 columns; default rate 11.144%.
- Missing cells: 1938; duplicates: 0; constants: 1.
- Near-constant features ($\geq 99.5\%$ one value): 1; 1e9-scale features: 0.

polish_bankruptcy

- Shape: 10503 rows × 65 columns; default rate 4.713%.
- Missing cells: 9888; duplicates: 87; constants: 0.
- Near-constant features ($\geq 99.5\%$ one value): 0; 1e9-scale features: 0.

south_german_credit

- Shape: 1000 rows × 21 columns; default rate 30.000%.
- Missing cells: 0; duplicates: 0; constants: 0.
- Near-constant features ($\geq 99.5\%$ one value): 0; 1e9-scale features: 0.

taiwan_bankruptcy

- Shape: 6819 rows × 96 columns; default rate 3.226%.
- Missing cells: 0; duplicates: 0; constants: 1.
- Near-constant features ($\geq 99.5\%$ one value): 2; 1e9-scale features: 23.

Best completed baseline results

dataset	protocol	model	balanced_accuracy	f1	roc_auc	average_precision
polish_bankruptcy	clean	random_forest	0.714286	0.459184	0.882141	0.420617
polish_bankruptcy	historical	random_forest	0.715784	0.473684	0.890819	0.426711
taiwan_bankruptcy	clean	xgb	0.821212	0.47619	0.952996	0.461308
pkdd_czech	clean	xgb	0.683607	0.48	0.808197	0.565598
pkdd_czech	historical	xgb	0.683607	0.48	0.824044	0.567113
taiwan_bankruptcy	historical	random_forest	0.771212	0.485437	0.943363	0.504721
south_german_credit	clean	svm	0.690476	0.566667	0.744405	0.595586
south_german_credit	historical	svm	0.697619	0.576271	0.751667	0.614627

Best completed TDA results

dataset	protocol	snapshot_variant	landmark_percent	model	balanced_accuracy	f1	roc_auc
south_german_credit	clean	revised	20	xgb	0.5	0.666667	0.7
south_german_credit	clean	historical500	20	xgb	0.504	0.668005	0.686302
pkdd_czech	clean	historical500	20	random_forest	0.562	0.69541	0.644464
pkdd_czech	clean	revised	20	xgb	0.9	0.909091	1
taiwan_bankruptcy	clean	revised	10	logistic	1	1	1
taiwan_bankruptcy	historical	revised	10	logistic	1	1	1
pkdd_czech	historical	revised	10	logistic	1	1	1
polish_bankruptcy	historical	revised	20	knn	1	1	1
polish_bankruptcy	clean	revised	10	random_forest	1	1	1
south_german_credit	historical	revised	10	random_forest	1	1	1

historical500	20	logistic	1	1	1	█
pkdd_czech	historical	historical500	20	xgb		
1	1	█	south_german_credit	historical	revised	
10	xgb	1	1	1	█	polish_bankruptcy
historical500	10	xgb	1	1	1	█
polish_bankruptcy	historical	historical500	20	xgb		
1	1	1	█	taiwan_bankruptcy	historical	historical500
20	random_forest	1	1	1	█	taiwan_bankruptcy
historical500	20	random_forest	1	1	1	

Best completed extended-experiment results

experiment	dataset	protocol	variant	landmark_percent	mode
setting		balanced_accuracy	f1	roc_auc	█
----- :----- :----- :----- :----- :-----					
----- :----- :----- :----- :----- :-----					█
16 pkdd_czech	clean	pca_component_sweep		nan	logistic
components=10		0.562295 0.219512		0.522951	█
16 polish_bankruptcy	clean	pca_component_sweep		nan	logistic
components=20		0.729243 0.222222		0.773671	█
13 pkdd_czech	clean	matched_pca_variance		nan	logistic
reference_variance=0.90		0.581148 0.24		0.551366	█
13 polish_bankruptcy	clean	matched_pca_variance		nan	logistic
reference_variance=0.90		0.736597 0.282353		0.797844	█
13 taiwan_bankruptcy	clean	matched_pca_variance		nan	logistic
reference_variance=0.90		0.854167 0.289062		0.932042	█
18 taiwan_bankruptcy	clean	pca_component_sweep		nan	logistic
components=5		0.878788 0.3083		0.928289	█
12 south_german_credit	clean	matched_sample_size		nan	logistic
t=12		0.6		0.333333 0.64	█
14 south_german_credit	clean	1_default_to_4_nondefault		10	logistic
20 default / 80 non-default snapshots		0.2625 0.337079		0.36	█
2 taiwan_bankruptcy	historical	tuned_baseline		nan	xgb
bounded_grid_3fold		0.781439 0.485981		0.952256	█
2 taiwan_bankruptcy	clean	tuned_baseline		nan	xgb
bounded_grid_3fold		0.782955 0.504854		0.948933	█
2 polish_bankruptcy	historical	tuned_baseline		nan	xgb
bounded_grid_3fold		0.751388 0.530612		0.918844	█
2 pkdd_czech	clean	tuned_baseline		nan	xgb
bounded_grid_3fold		0.737978 0.533333		0.832787	█
2 polish_bankruptcy	clean	tuned_baseline		nan	xgb
bounded_grid_3fold		0.766789 0.555556		0.91622	█
2 pkdd_czech	historical	tuned_baseline		nan	xgb
bounded_grid_3fold		0.721038 0.56		0.883607	█
2 south_german_credit	historical	tuned_baseline		nan	svm
bounded_grid_3fold		0.685714 0.565217		0.764405	█
2 south_german_credit	clean	tuned_baseline		nan	svm
bounded_grid_3fold		0.689286 0.569343		0.765893	█
13 south_german_credit	clean	matched_pca_variance		nan	logistic
reference_variance=0.90		0.692857 0.575342		0.74869	█
18 south_german_credit	clean	pca_component_sweep		nan	logistic
components=20		0.696429 0.57931		0.740476	█
12 taiwan_bankruptcy	clean	matched_sample_size		nan	logistic
t=6		0.6875 0.615385		0.84375	█
4 south_german_credit	clean	historical500		20	logistic
default		0.508 0.669355		0.871544	█
6 south_german_credit	clean	historical500		20	logistic
default		0.533 0.680356		0.889836	█
14 pkdd_czech	clean	1_default_to_4_nondefault		10	logistic
20 default / 80 non-default snapshots		0.5375 0.683761		0.955	█
19 south_german_credit	clean	revised		10	linear_regression
default		0.689655 nan		19 pkdd_czech	clean
revised		20 linear_regression		default	
0.6		0.714286 nan		11 pkdd_czech	clean
revised		20 logistic		default	
0.6		0.714286 1		11 south_german_credit	clean
revised		20 logistic		default	
0.6		14286 1		6 pkdd_czech	clean
revised		10 logistic		default	
0.8		33 1		12 pkdd_czech	clean
matched_sample_size		nan logistic		t=6	
0.833333 0.857143		1		12 polish_bankruptcy	clean
matched_sample_size		nan logistic		t=12	
0.888889 0.888889		0.938272		14 polish_bankruptcy	clean
1_default_to_4_nondefault		10 logistic		20 default / 80 non-default snapshots	
0.8875 0.898876		1		4 pkdd_czech	clean
revised		20 xgb		default	
0.9		11 polish_bankruptcy		clean	historical500
		logistic		default	
0.945 0.947867		1			

6	polish_bankruptcy	clean	historical500	20	
ogistic	default		0.95	0.952381	1
14	taiwan_bankruptcy	clean	1_default_to_4_nondefault	10	logi
stic	20 default / 80 non-default snapshots		0.975	0.97561	1
19	south_german_credit	historical	revised	20	linear_
regression	default		1	1	nan
11	south_german_credit	historical	revised	10	logistic
	default		1	1	
6	south_german_credit	historical	revised	10	logistic
	default		1	1	
4	south_german_credit	historical	revised	10	logistic
	default		1	1	
	taiwan_bankruptcy	clean	revised	10	logistic
	default		1	1	
	aiwan_bankruptcy	clean	revised	10	logistic
	fault		1	1	
sh_bankruptcy	clean	revised		10	linear_regression
lt		1	1	nan	
bankruptcy	historical	historical500		20	logistic
		1	1	1	
krupcty	historical	historical500		20	linear_regression
		1	1	nan	
ptcy	historical	historical500		20	logistic
		1	1	1	
y	historical	historical500		10	logistic
		1	1	1	
	historical	historical500		10	linear_regression
		1	1	nan	
historical	historical500		10	logistic	default
		1	1	1	
torical	historical500		20	random_forest	default
		1	1	1	
	historical500		20	xgb	default
		1	1	1	
l	historical500		10	logistic	default
		1	1	1	
historical500			20	logistic	default
		1	1	1	
vised			20	linear_regression	default
		1	1	nan	
ed			20	logistic	default
		1	1	1	
al500			20	logistic	default
		1	1	1	
00			20	linear_regression	default
		1	1	nan	
			10	logistic	default
1		1	1	1	

Limitations

- Mirror provenance and licence text remain unresolved against primary records.
- Independent snapshot classification can be degenerate when train/test snapshot distributions are nearly indistinguishable.
- Revised snapshot counts implement the Experiment 24 calibration $l = \text{ceil}(n_{\text{class}} / t)$ separately by spl it and landmark size.

- Historical500 status at report generation: 16/16 configurations complete; see 6_Results/New_Datasets/active_jobs.json.

Reproduction

powershell

```
.\tda_env\Scripts\python.exe run_new_datasets.py --stages ingest baseline tda report
.\tda_env\Scripts\python.exe run_remaining_experiments.py
.\tda_env\Scripts\python.exe run_remaining_experiments.py --experiments 4 6 7 8 10 11 15 19 25 27
.\tda_env\Scripts\python.exe run_new_datasets.py --stages report
```